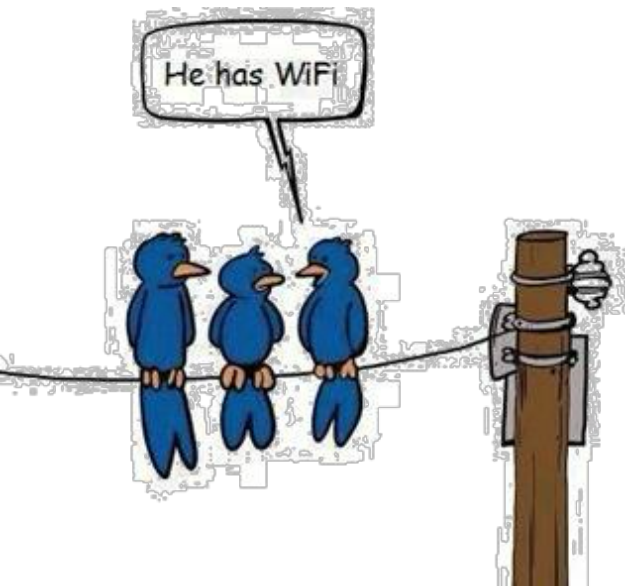


ECSE 4560/ 6560

Modern Communication Systems

Lecture 23: Wireless Sensing Part 4: UWB and FMCW Intro

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **FMCW**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Final Project Presentation and Demos: **Fri 12/12**
 - Project Report and peer evaluation: Finals exam week
 - Instructions will be posted on Piazza
- HW/Exams Dates
 - HW 6, HW 7 and Exam2 will be graded soon
 - HW 8 will be posted; due next week



Today's Class

- **Wireless Sensing**

- Recall: ToF-based WiFi localization
- UWB-based localization 
- FMCW Radars 
- RangeFFT
- Range Resolution in FMCW Radar
- AoA-FFT
- Doppler-FFT

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



- How about packet detection delay & processing delay?
 - Use OFDM to correct for packet detection delay
 - Estimate and calibrate for processing delay

Not Practical!

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- Resolution limited by sampling rate (bandwidth)!

Multipath problem

$$\boxed{\Delta d = \Delta \tau \times c} \approx 1/B$$

$$802.11n \text{ bandwidth} = 40MHz \Rightarrow \Delta \tau = 25ns \Rightarrow \boxed{\Delta d = 12.5 m}$$

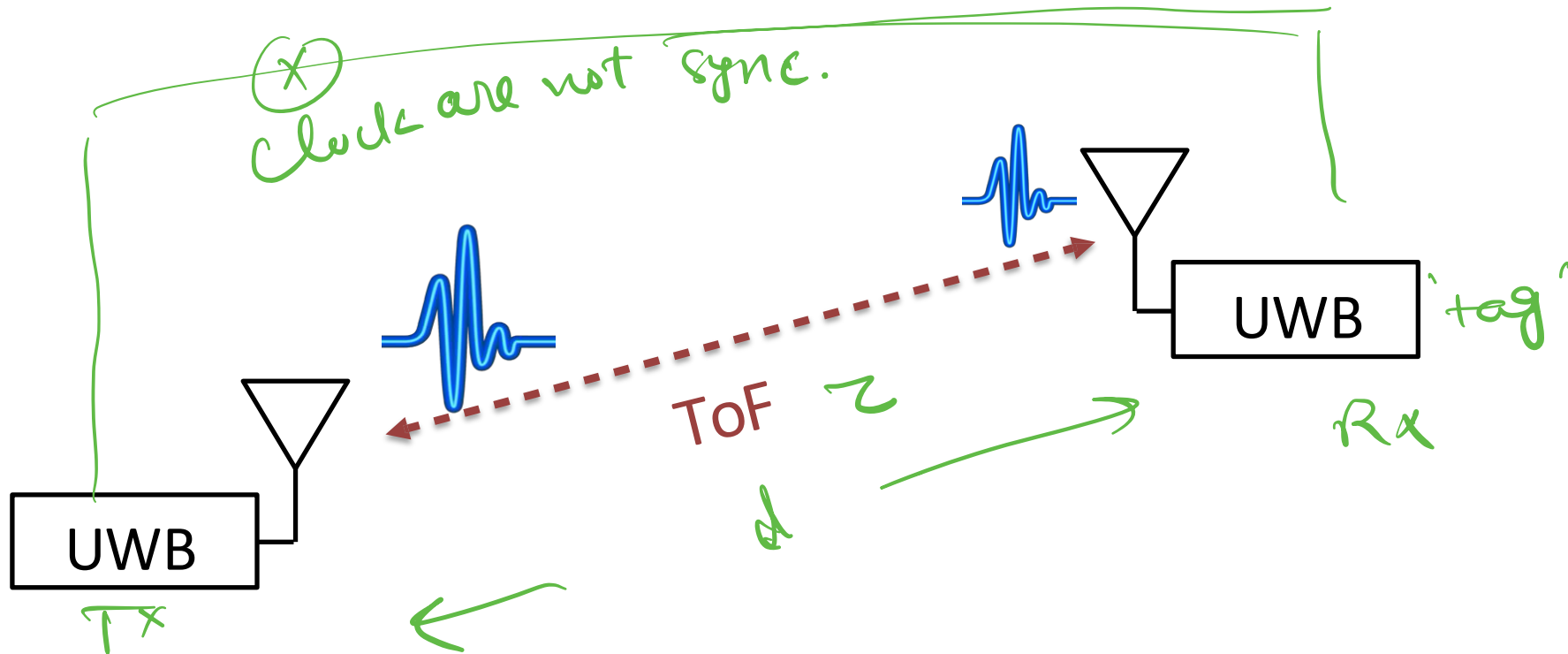
- Other systems than WiFi can get accurate ToF:

Limited resolution

- UWB: Ultra-Wide Band
- FMCW: Frequency Modulated Carrier Wave

Not Supported in WiFi
(Will discuss in ~~future~~ lectures)
today

UWB: Ultra-Wide Band



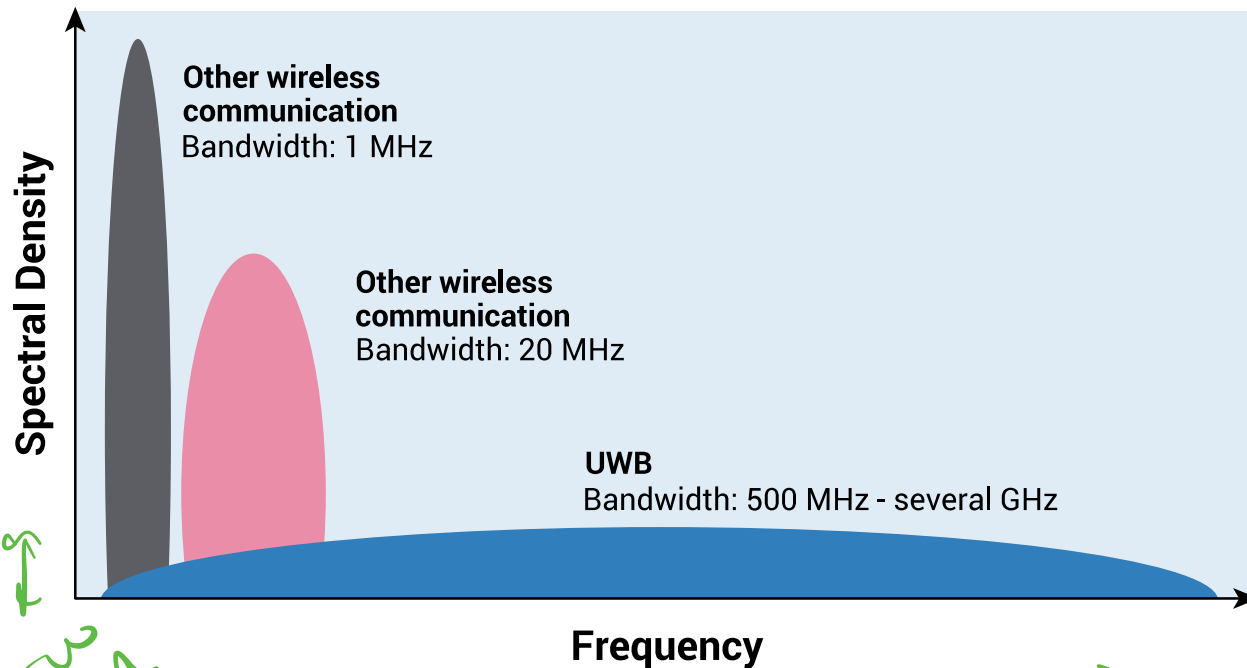
Localize by measuring the Time-of-flight

Distance = Time-of-flight × speed of light

$d = z \times c$

UWB: Ultra-Wide Band

not for comm.
but for ranging / sensing

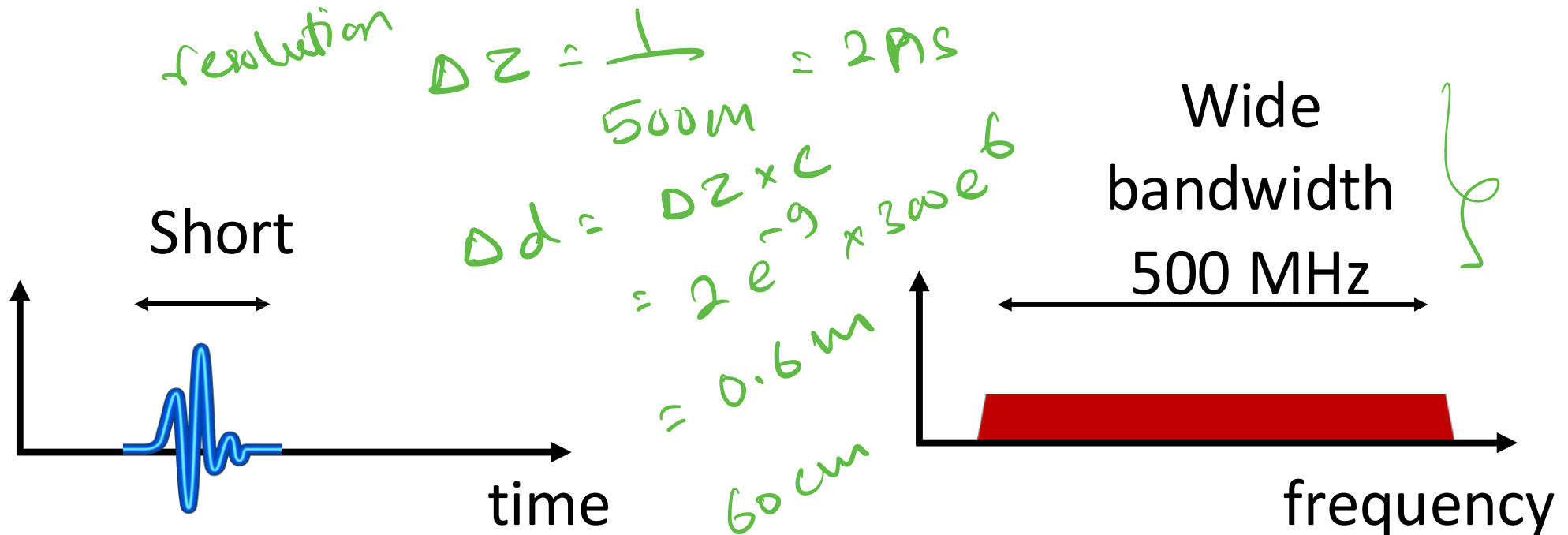


- UWB transmits extremely short pulses or wideband signals, spreading energy across a large spectrum.
- Part of IEEE standard 802.15.4 standards
- Co-exists with other technologies at 3.1–10.6 GHz band

low power
> 500 MHz

UWB: Ultra-Wide Band

Short pulse allows measuring time at very fine granularity

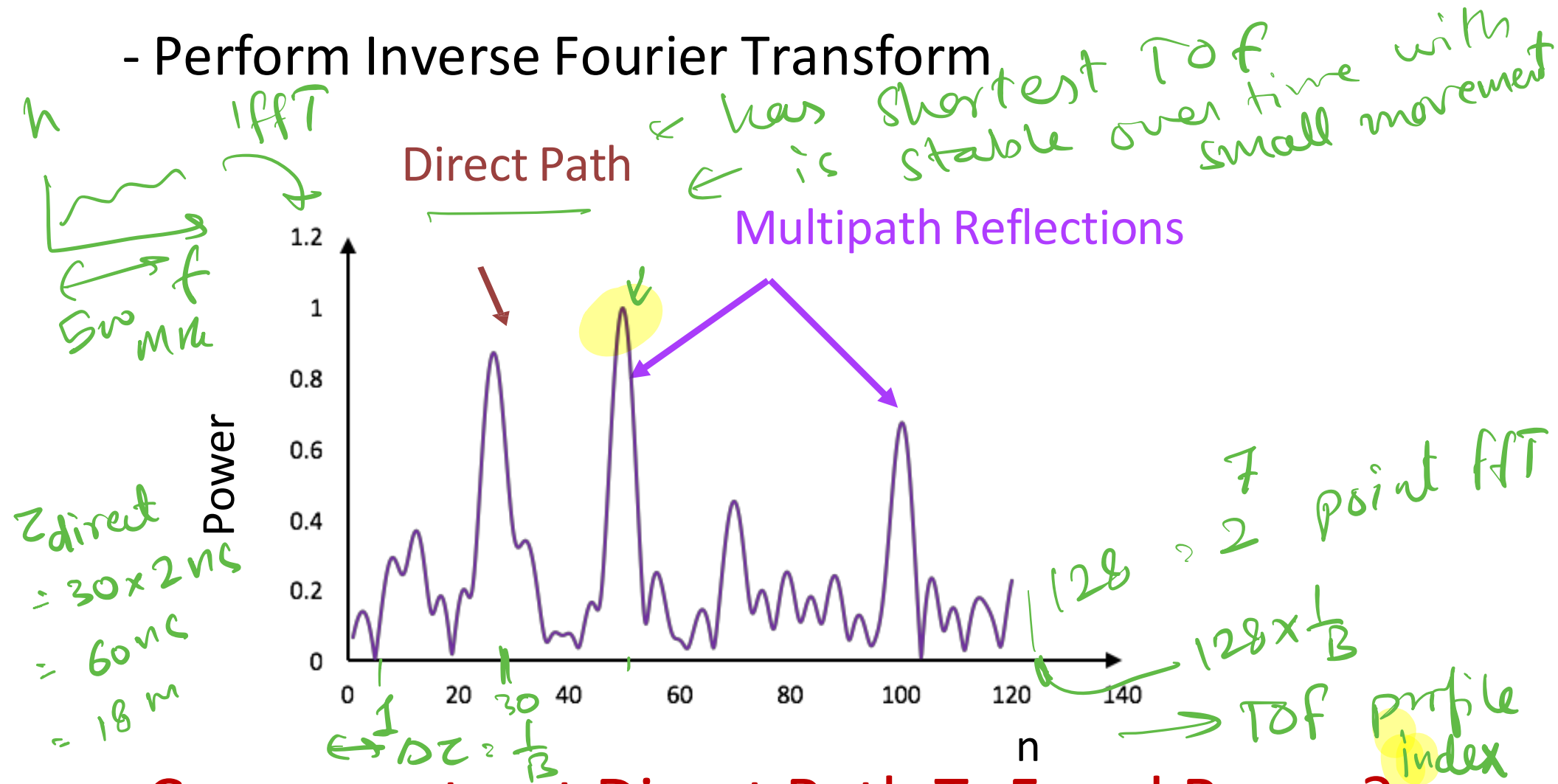


How wide bandwidth helps with Range/TOF estimation?

Estimating the Time-of-Flight

- Wide bandwidth can be used to estimate the channel taps in the time domain

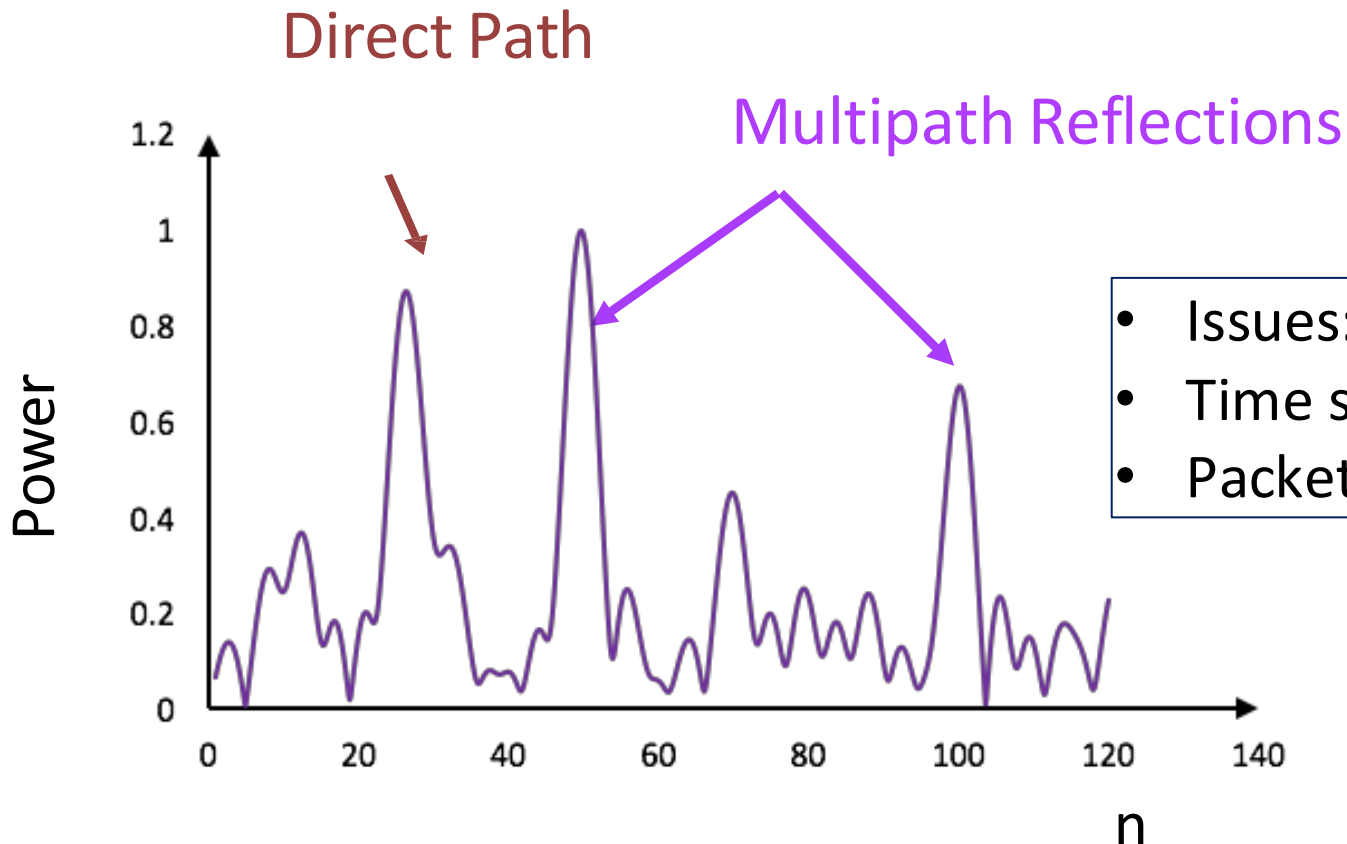
- Perform Inverse Fourier Transform



Can we extract Direct Path ToF and Range?

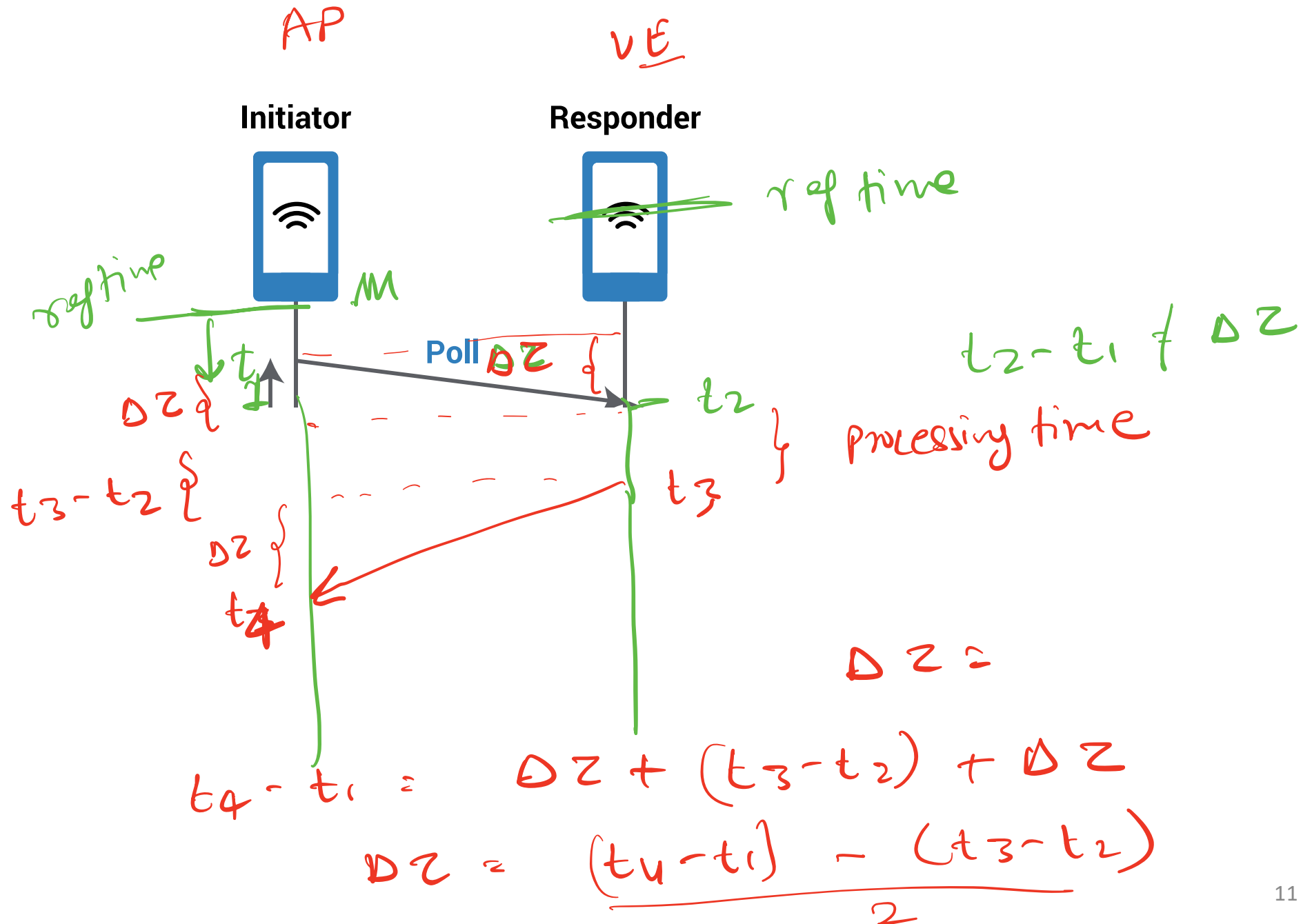
Estimating the Time-of-Flight

- Wide bandwidth can be used to estimate the channel taps in the time domain
 - Perform Inverse Fourier Transform

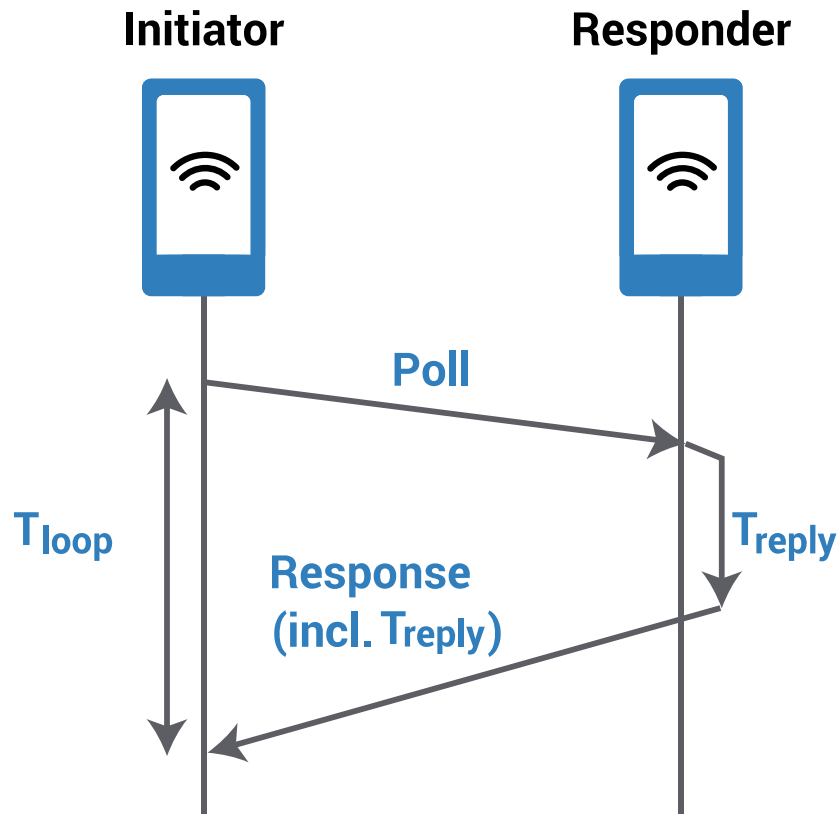


Can we extract Direct Path ToF and Range?

Two-Way Ranging (TWR) in UWB



Two-Way Ranging (TWR) in UWB

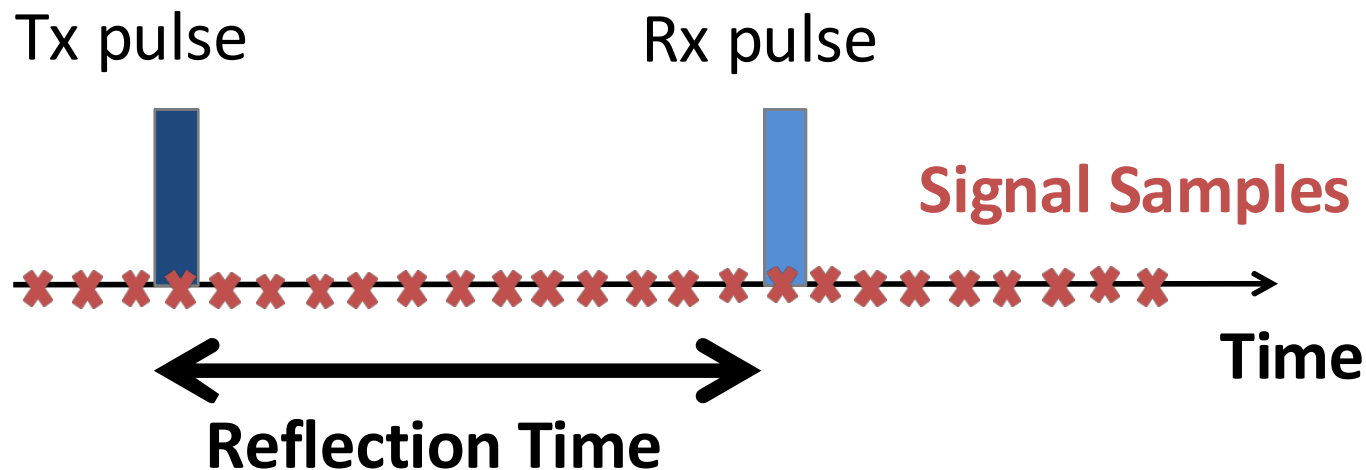


- Use TWR for coarse ToF estimation
- Use multipath profiling for fine-grained refining

absolute
⇒ combine with multi-antenna ⇒ good accuracy for localization

Measuring Reflection Time

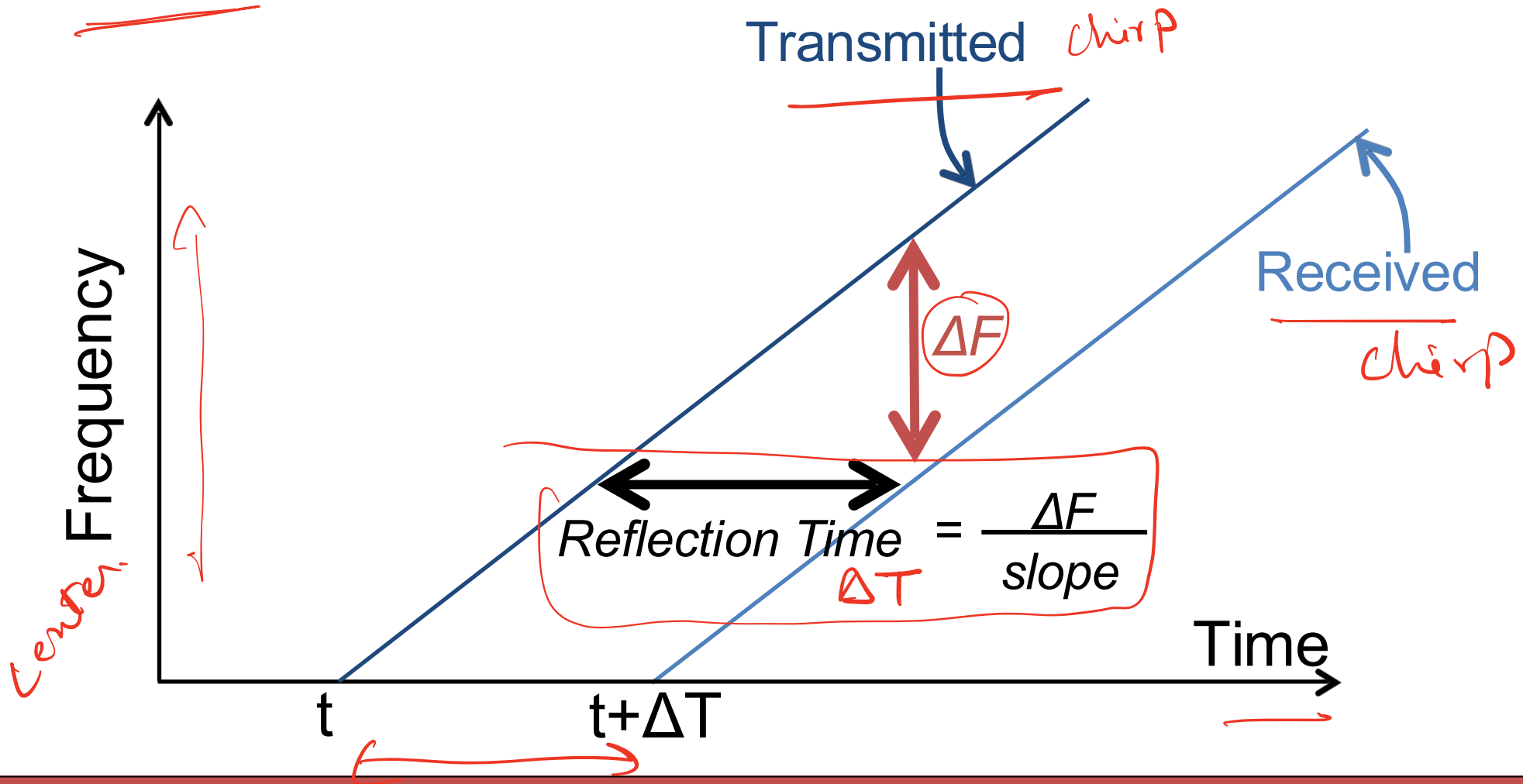
- Option1: Transmit short pulse and listen for the echo.



Need to sample at very high rate : UWB

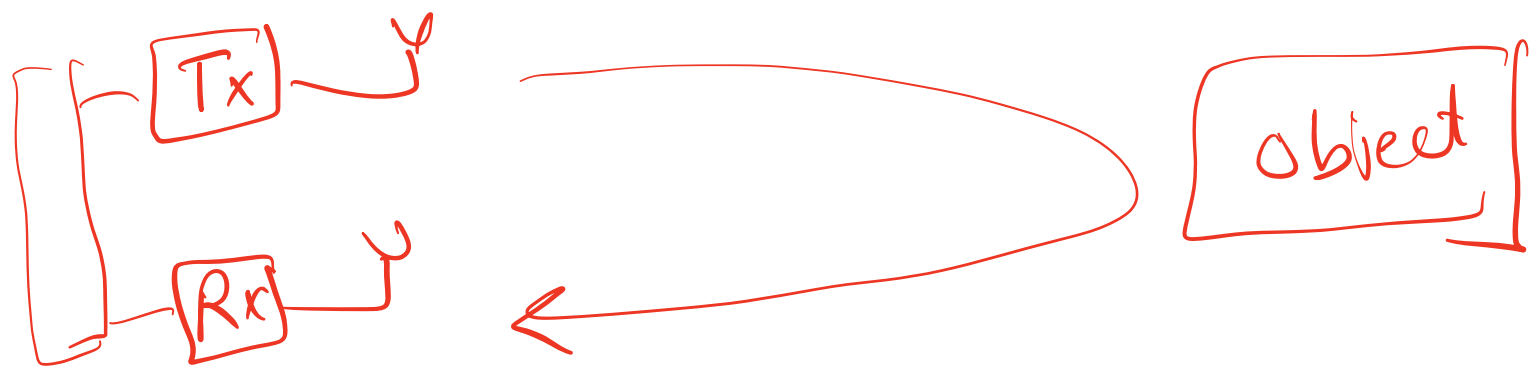
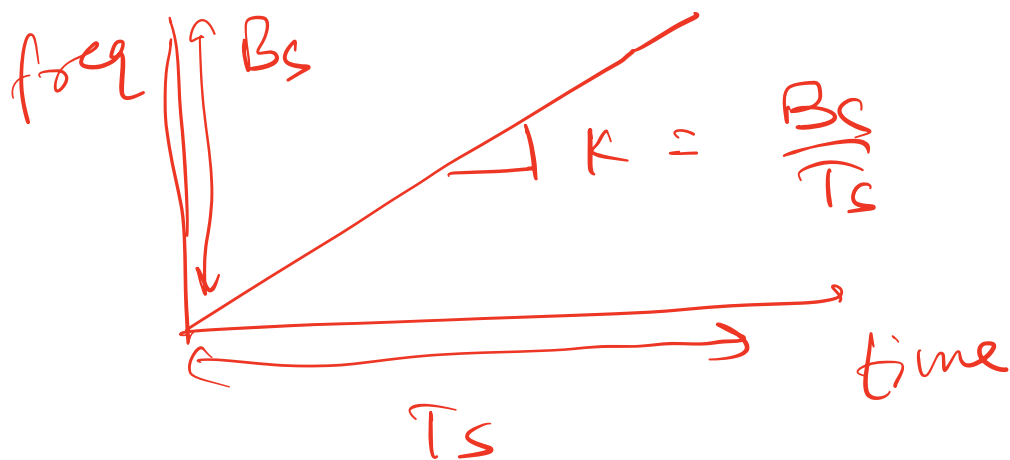
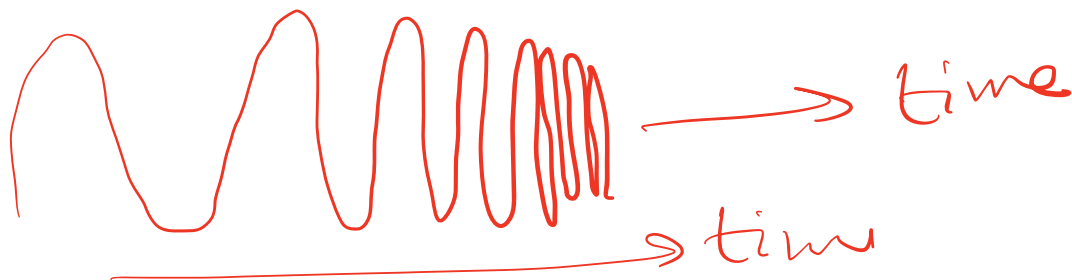
Multi-GHz samplers are expensive and generate high noise: not suitable for this application

FMCW: Measure time by measuring frequency



How do we measure ΔF ?

T_x



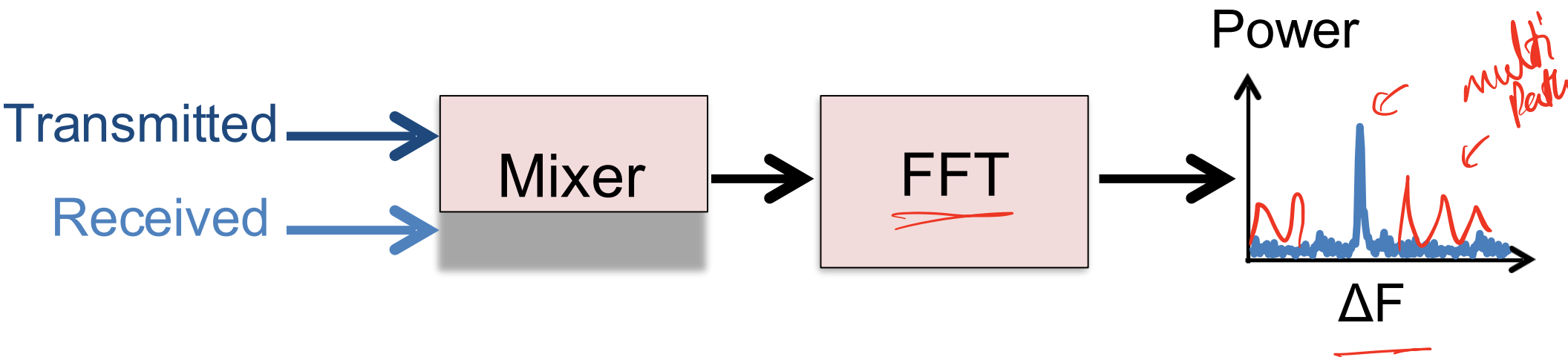
T_x & R_x are co-located $\downarrow$

(i) No issue of clock sync

(ii) No need of GHz sampling
beacuz sweeping center freq.
LO (local oscillator)

Measuring ΔF

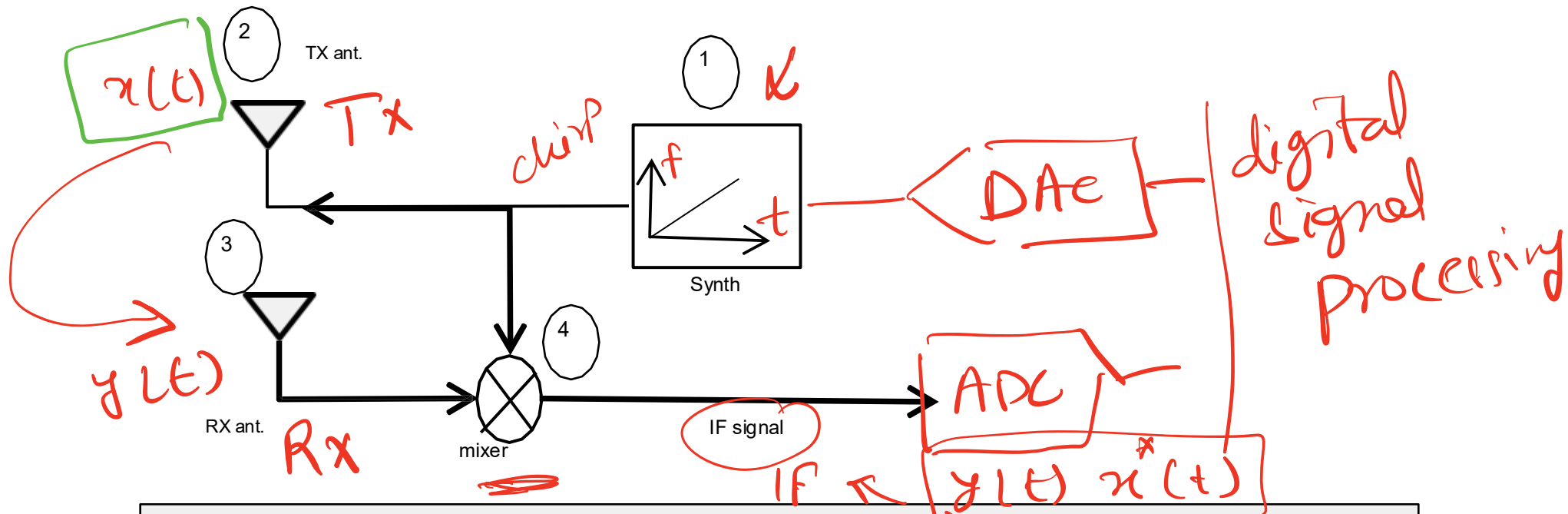
- Subtracting frequencies is easy (e.g., removing carrier in WiFi)
- Done using a mixer (low-power; cheap)



Signal whose frequency is ΔF

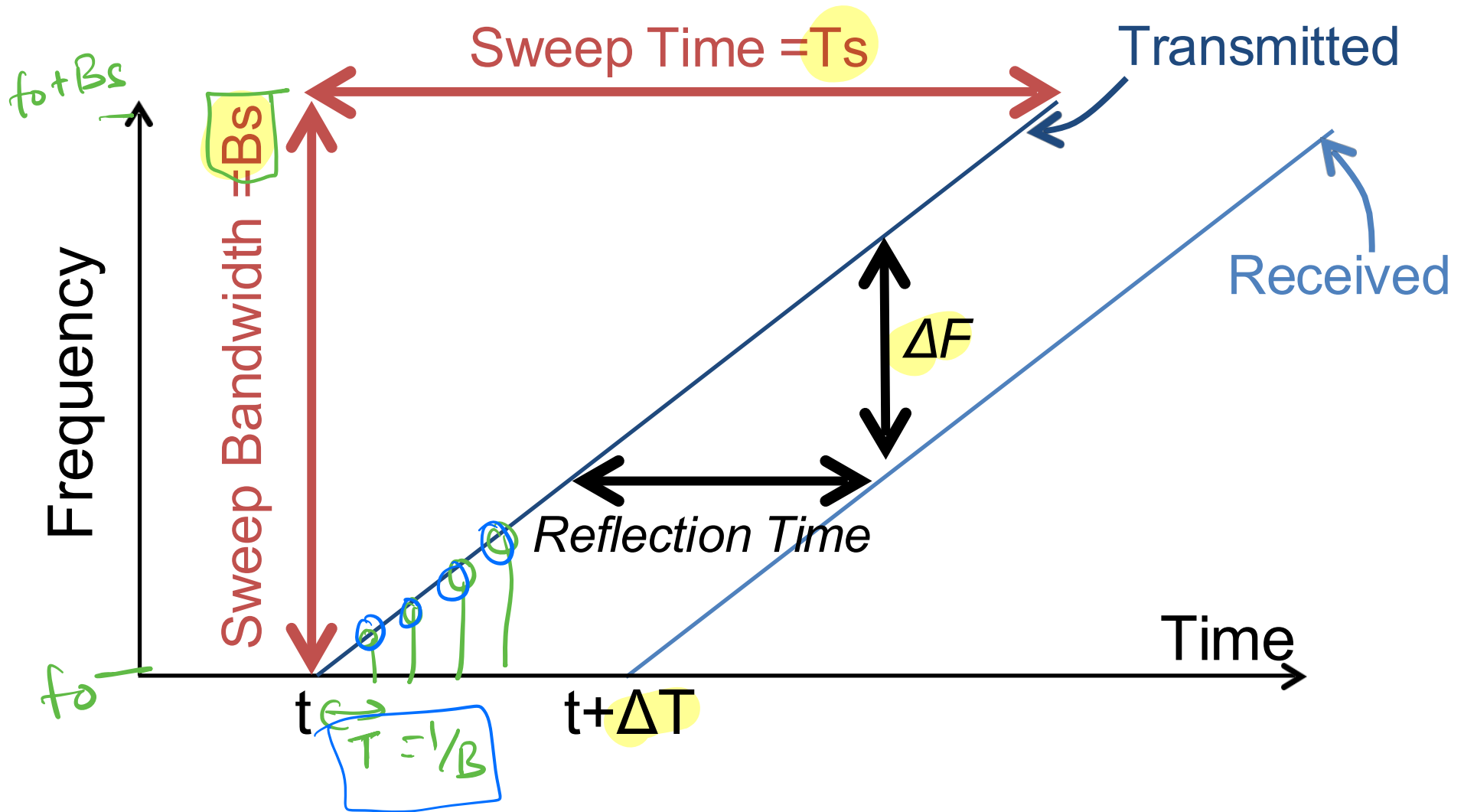
$\Delta F \rightarrow$ Reflection Time $\rightarrow$ Distance

FMCW Radar Architecture



1. A synthesizer (synth) generates a chirp
2. The chirp is transmitted by the TX antenna
3. The chirp is reflected off an object and the reflected chirp is received at the RX antenna.
4. The RX signal and TX signal are 'mixed' and the resulting signal is called an 'IF signal'. We'll analyze the IF signal in more detail in the next slide

FMCW: Measure time by measuring frequency



Slope = $k = B_s/T_s$

Reflection Time = $\Delta F/k$

FMCW

def'n of freq. $\downarrow$ phase is $\propto t^2$

- FMCW Transmitted Signal

$$x(t) = e^{j2\pi(\frac{k}{2}(t^2 + f_0 t))}$$

slope

Frequency is linear in time;
hence phase is quadratic

f_0 = start freq
 $f_0 + B_s$ = end freq

$$\text{freq} = Kt + f_0$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2}((t-\tau_i)^2 + f_0(t-\tau_i)))}$$

reflector i

delay τ_i

Reflections linearly
combine over the
wireless medium

nonconversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$

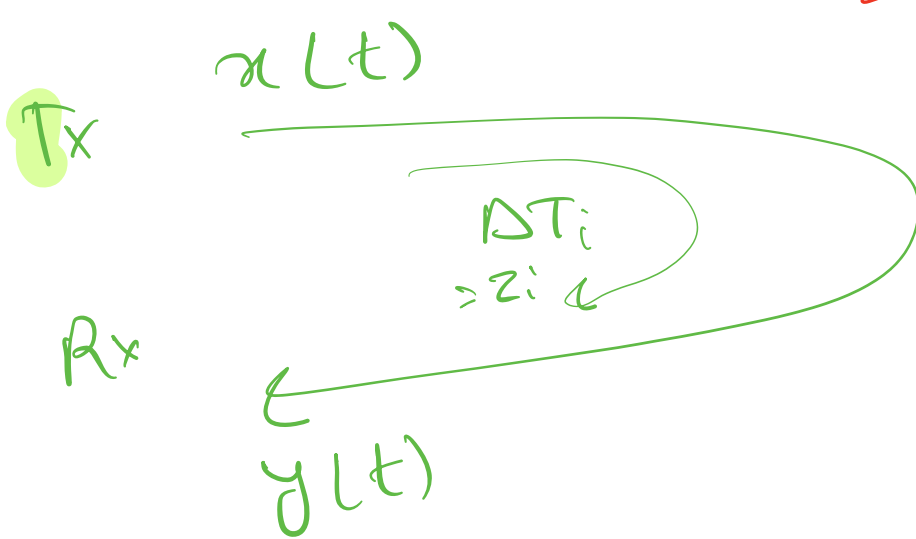
frequency $k\tau_i$

find peak freq of $y_b(t)$

$$y(t) x^*(t)$$

$+ z_i^2 \frac{k}{2} \rightarrow \text{ignore}$

$$\frac{k}{2} \tau_i^2 \ll z_i^2 f_0$$



$$y(t) = \sum_i A_i x(t - \Delta T_i)$$

$$= \sum_i A_i e^{j2\pi \left(\frac{\kappa}{2} (t - \Delta T_i)^2 + f_0(t - \Delta T_i) \right)}$$

$$= \sum_i A_i e$$

$$y_b(t) = y(t) x^*(t)$$

$$= \sum_i A_i e^{j2\pi \left(\frac{\kappa}{2} (t - z_i)^2 + f_0(t - z_i) \right)}$$

$$= \sum_i A_i e^{j2\pi \left(\frac{\kappa}{2} t^2 + f_0 t \right)}$$

$$= \sum_i A_i e^{j2\pi \left(\frac{\kappa}{2} \left[(t - z_i)^2 - t^2 \right] + f_0 \left[t - z_i - t \right] \right)}$$

$$= \sum_i A_i e^{j2\pi \left(\frac{\kappa}{2} \left[-2tz_i + z_i^2 \right] - f_0 z_i \right)}$$

FMCW

- FMCW Transmitted Signal:

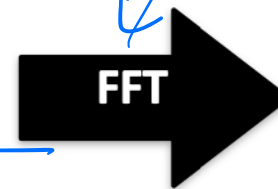
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

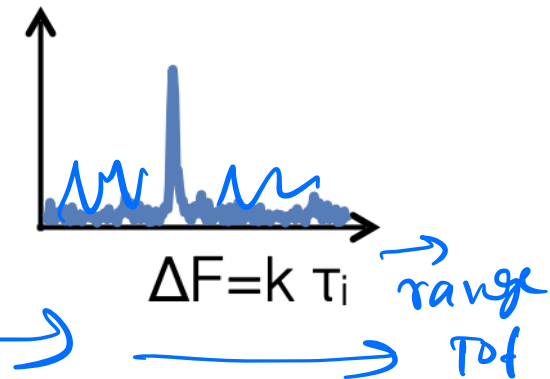
$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

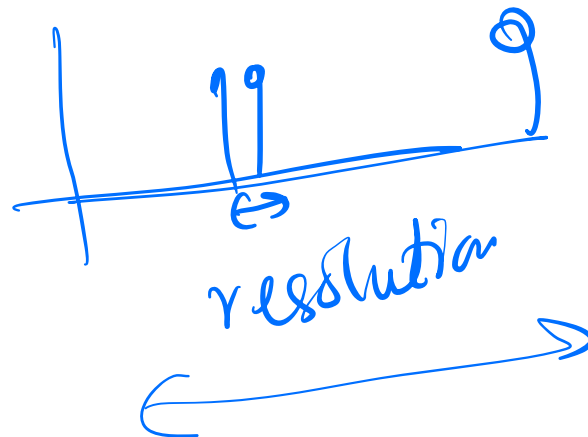
$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$



Range FFT



↑ range
is
embedded in
freq
domain



FMCW Max Range

- FMCW Transmitted Signal:

$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

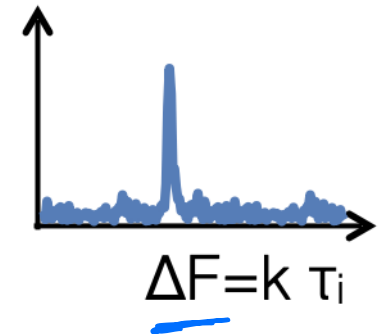
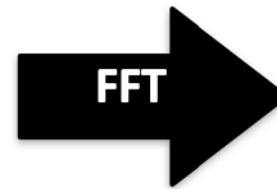
- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

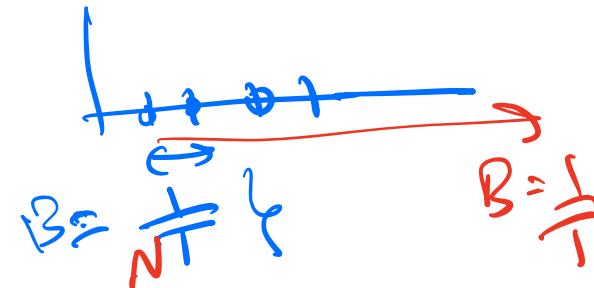
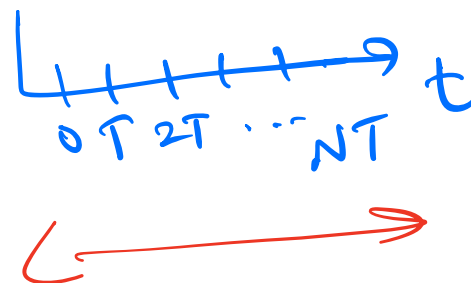
$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$

$\uparrow$
 nT



- Sampling Rate = B

$$\Delta F < B \xrightarrow{\text{max}} \tau_{\max} = B/k = B \times T_s / B_s \xrightarrow{\text{max}} d_{\max} = c \times B \times T_s / 2 B_s$$



FMCW Range Resolution

- FMCW Transmitted Signal:

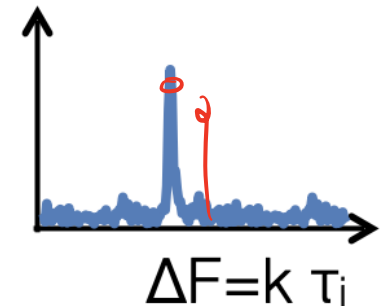
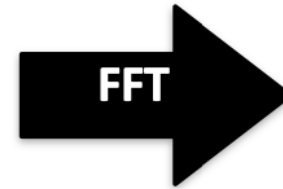
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$



- Sampling Rate = B

$$\Delta F < B \longrightarrow \tau_{\max} = B/k = B \times T_s / B_s \longrightarrow d_{\max} = c \times B \times T_s / 2 B_s$$

- Sampling Window = T_s

$$dF > 1/T_s \longrightarrow \tau_{\min} = 1/(k \times T_s) = 1/B_s \longrightarrow d_{\min} = c/2B_s$$

FMCW Range Estimation Summary

- FMCW Transmitted Signal:

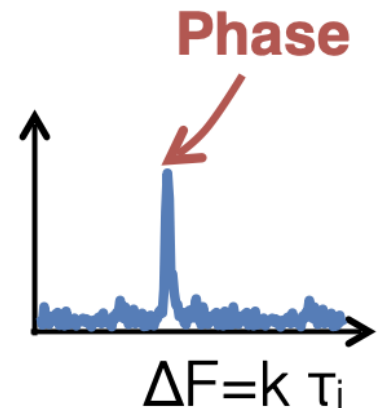
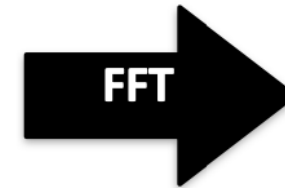
$$x(t) = e^{j2\pi(\frac{k}{2} t^2 + f_0 t)}$$

- FMCW Received Signal:

$$y(t) = \sum_i A_i e^{j2\pi(\frac{k}{2} (t-\tau_i)^2 + f_0(t-\tau_i))}$$

- FMCW after downconversion:

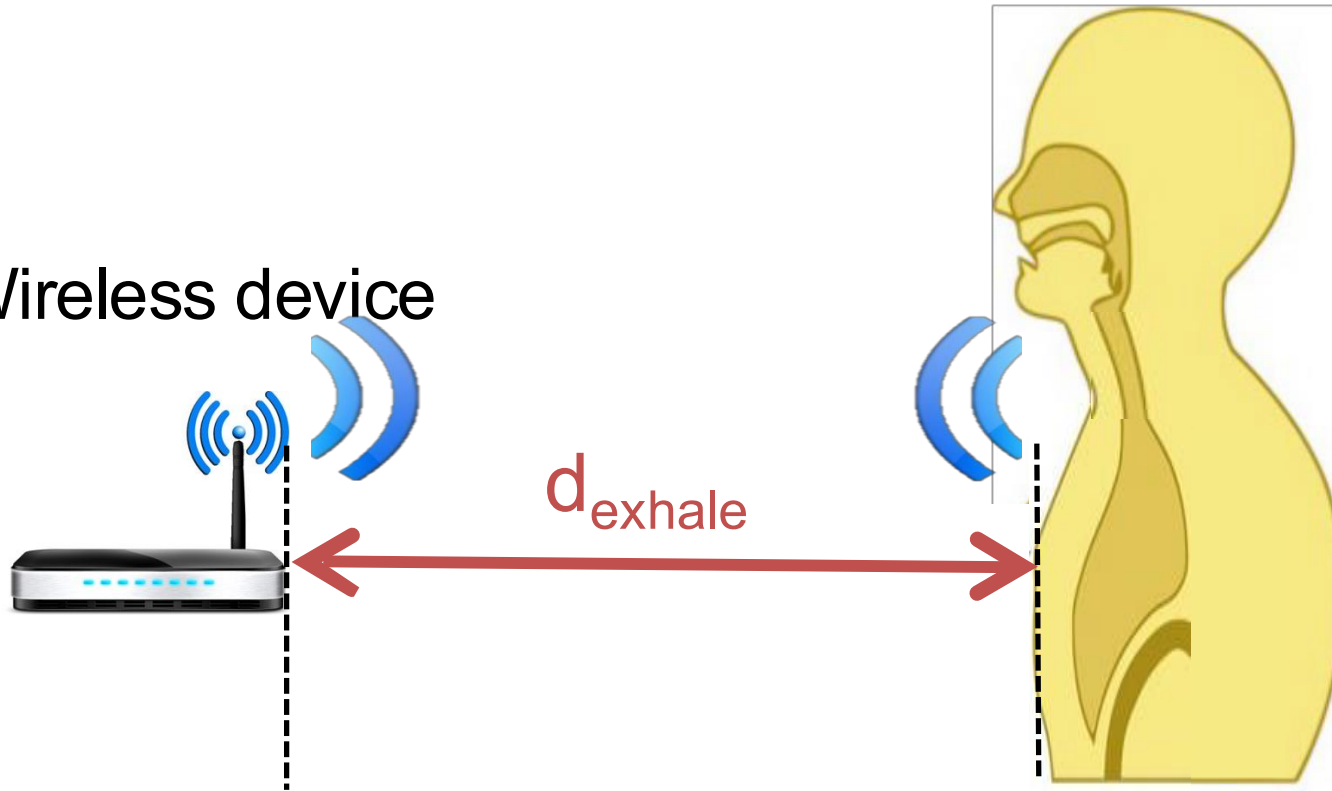
$$y_b(t) = \sum_i A_i e^{j2\pi(k\tau_i t + f_0\tau_i)}$$



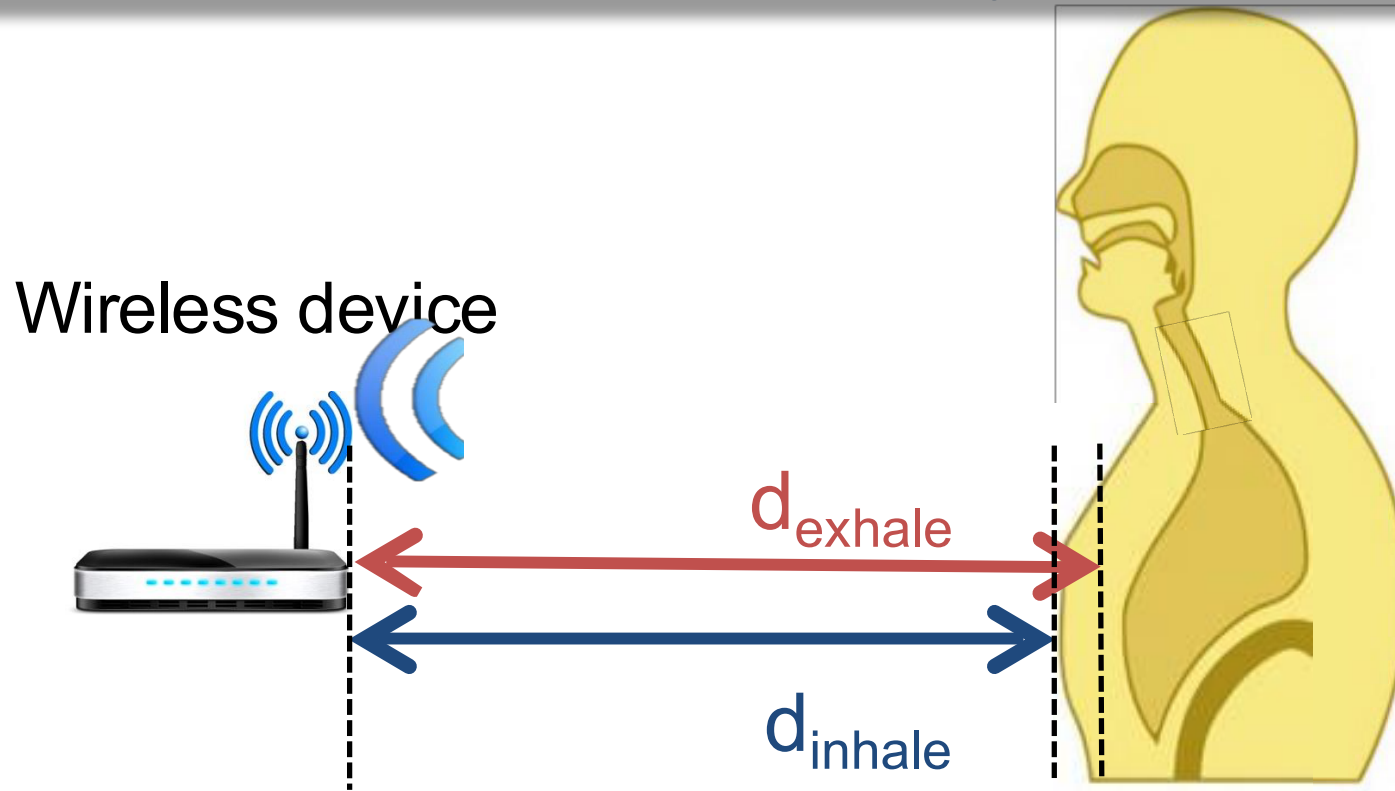
- Phase of peak = $f_0\tau_i$
 - Phase wraps around 2π
 - Use peak position $\Delta F = k \tau_i$ for course estimate of τ_i
 - Use peak phase $f_0\tau_i$ for fine estimate of τ_i

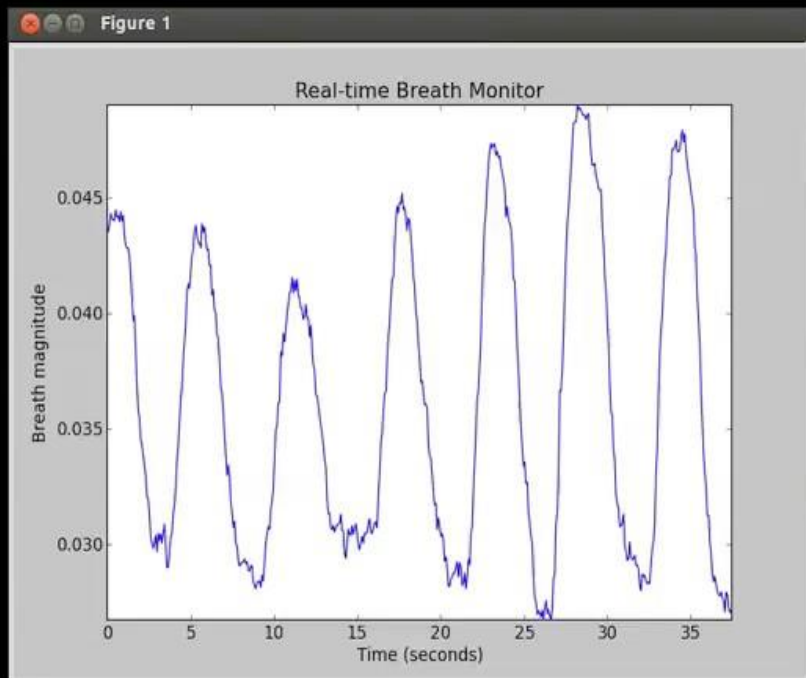
Vital Radio: Use wireless reflections off the human body to monitor breathing and heart rate

Wireless device



Problem: Localization accuracy is only 12cm and cannot capture vital signs





Let's zoom in on these signals

